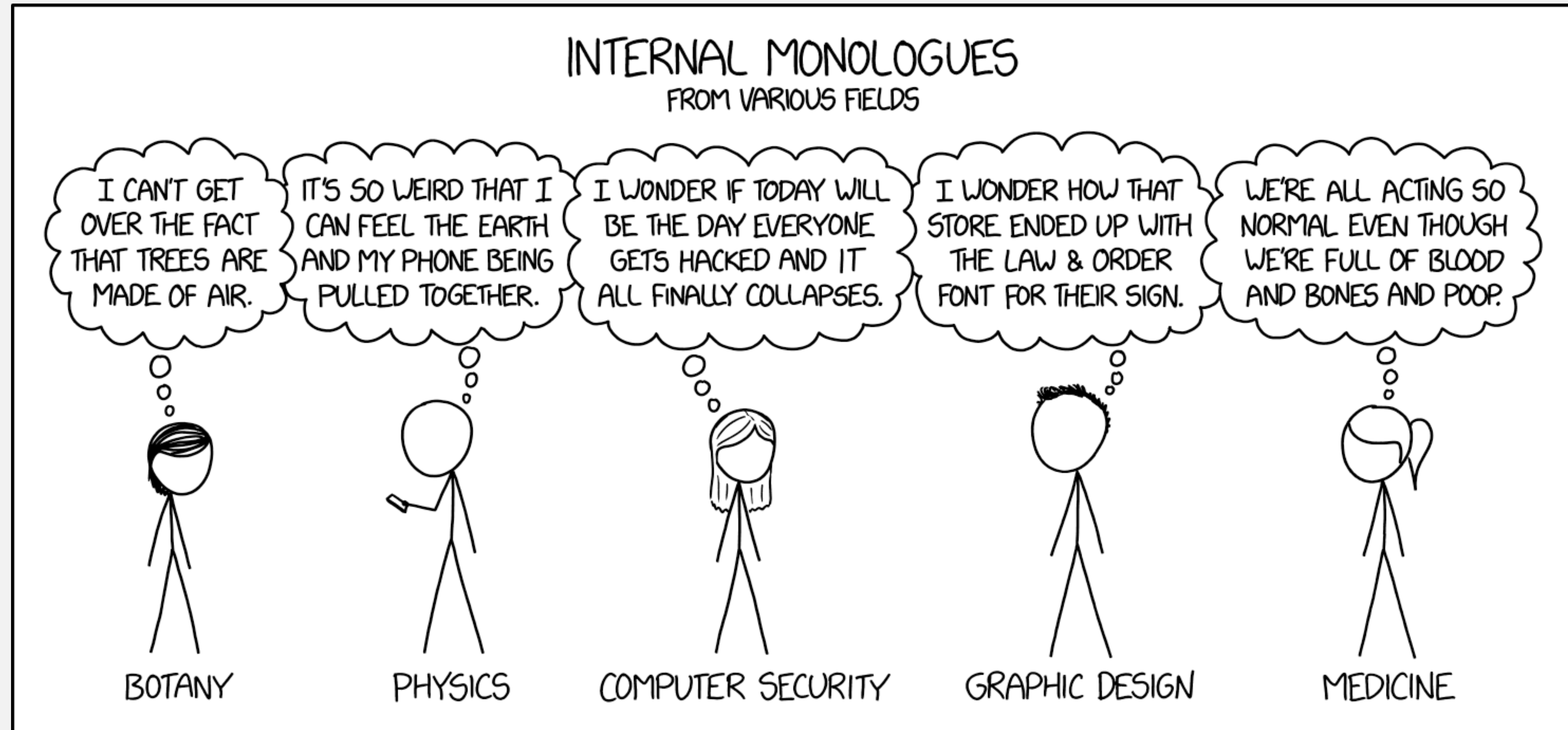


Organisational Infos



The Team

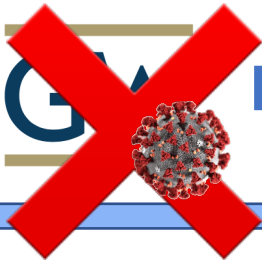
Peter Mayer



SDU

KIT

GW

Prof	
Postdoc	  
Dissertation	 
Studies	 

SDU

KIT

GW

My Research

- ▶ Useful
- ▶ Easy to use
- ▶ Fun to use
- ▶ ...

- ▶ Security Mechanisms & Interfaces
- ▶ Awareness, Education, & Training

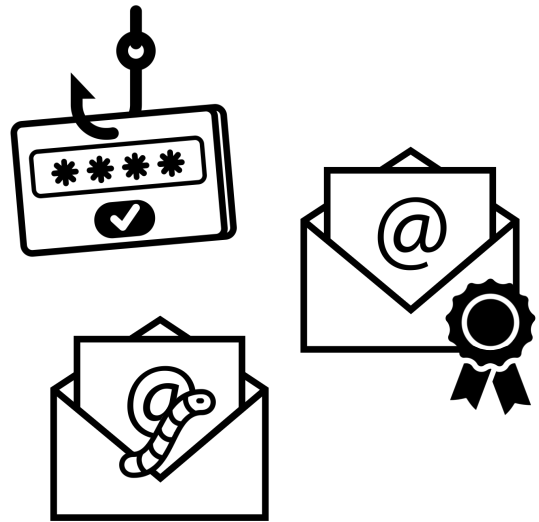
End-user Viable Information Security and Privacy Solutions

- ▶ Lay people
- ▶ Developers
- ▶ Decision Makers
- ▶ ...

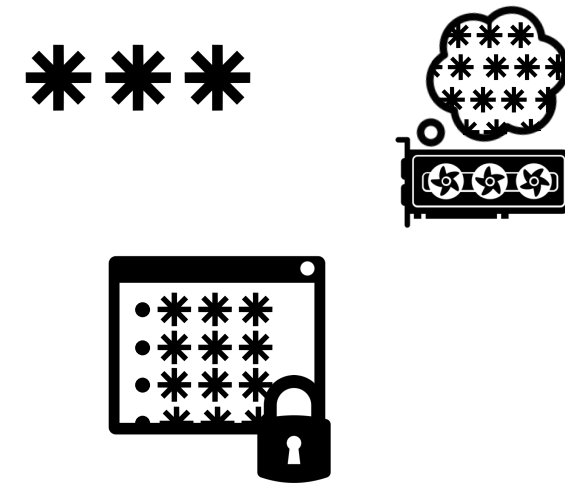
- ▶ Traditional Cybersecurity
- ▶ Privacy
- ▶ Informational Self-Determination
- ▶ ...

My Research

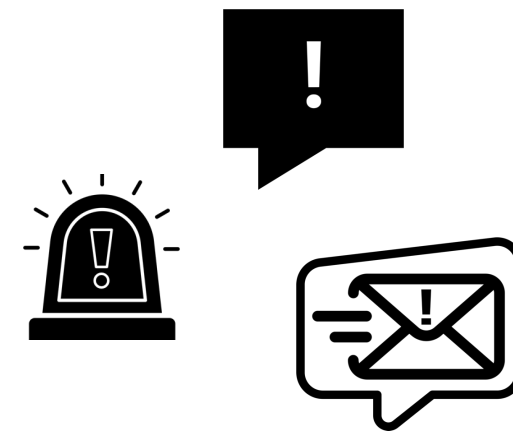
Email Security



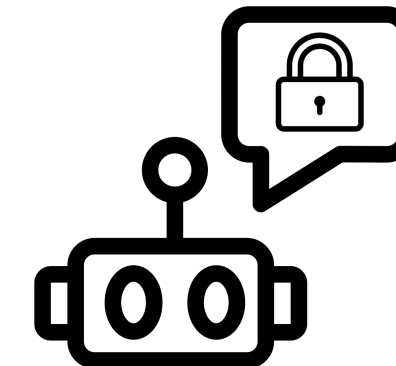
Usable & Secure Password Managers



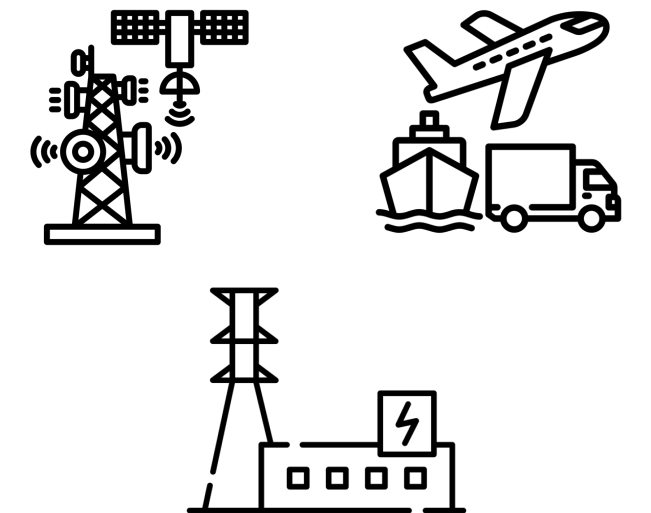
Security Notifications



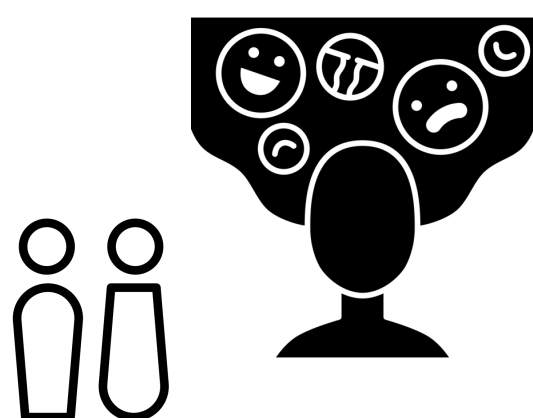
Security and Privacy Aspects of AI



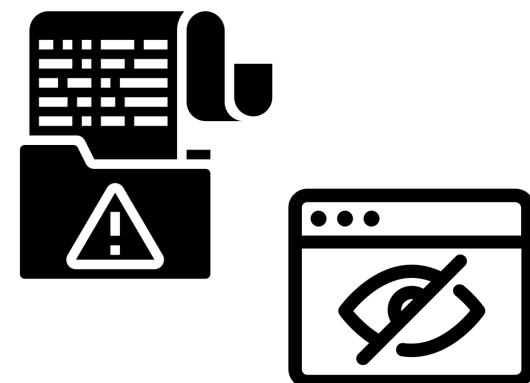
Critical Infrastructure



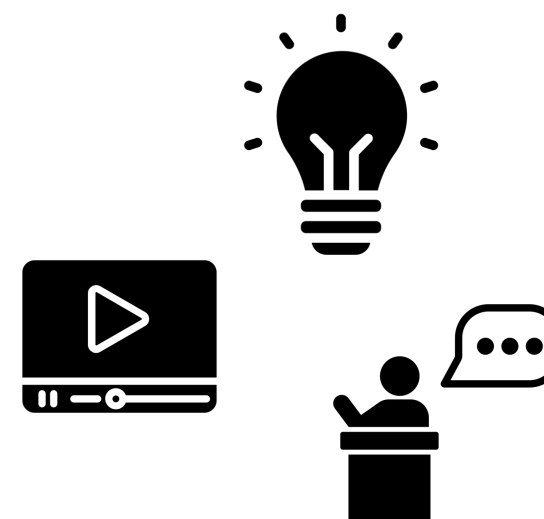
Demographic Effects in Security



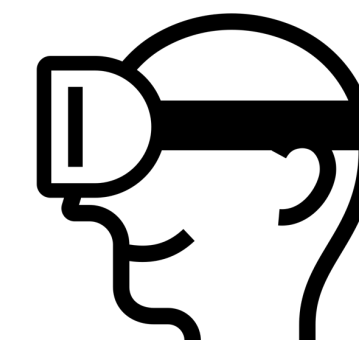
Privacy



Security Awareness & Education



Augmented & Virtual Reality



Supply Chain Security



Safran Thestrup Preisel

Academic

- ▶ BSc in Computer Science at SDU Odense
- ▶ Currently pursuing a MSc in Computer Science at SDU Odense

Interested in

- ▶ Digital Infrastructure
- ▶ Algorithms

Who are you?

Why this lecture?

Why learn about networking?

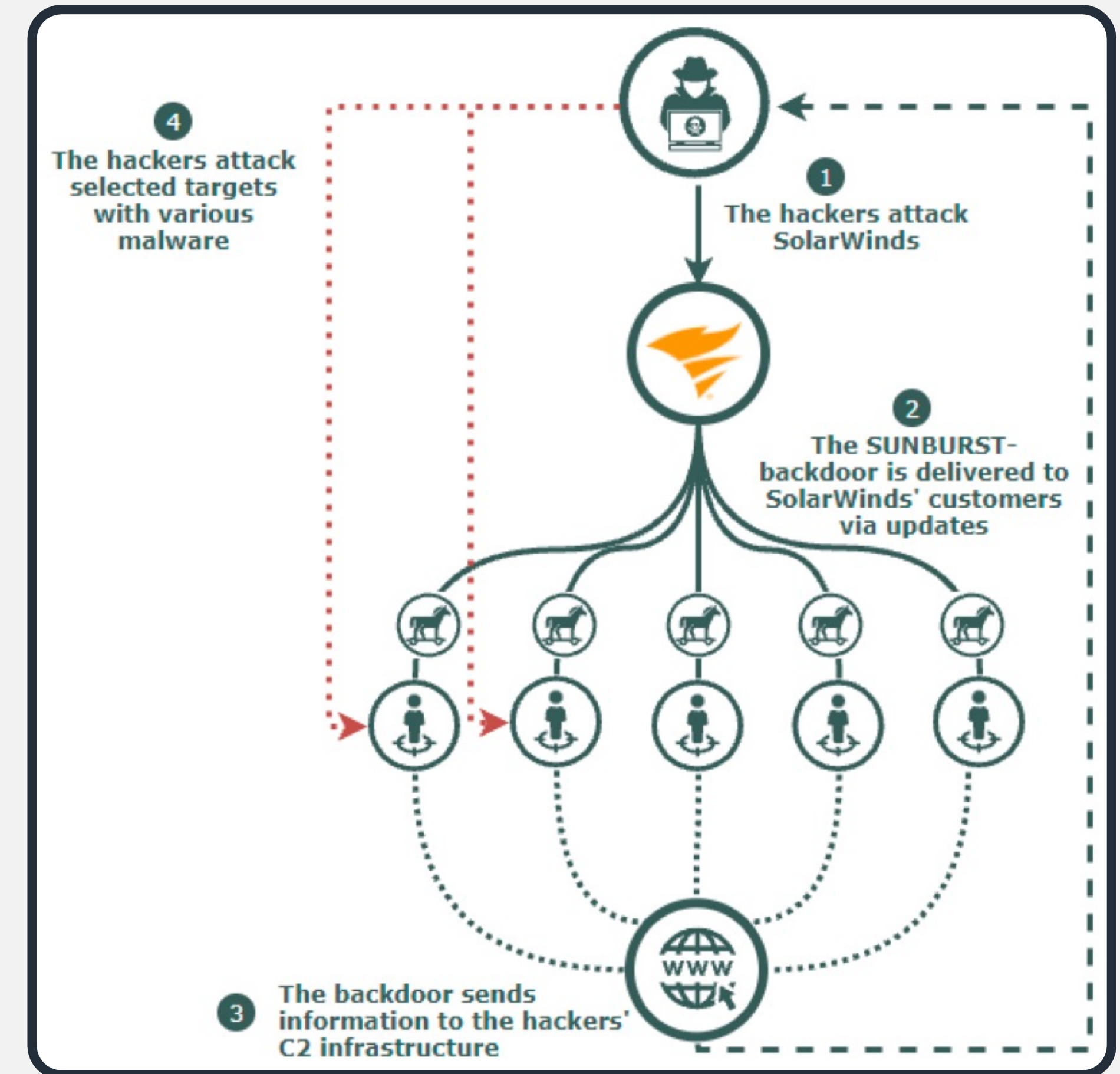


Why learn about cybersecurity?



Why learn about cybersecurity?

- ▶ SolarWinds
 - ▶ Develops widely used IT network management system
 - ▶ Used by high-profile targets
- ▶ The attack
 - ▶ Hackers got access to the internal network (how is still unclear!)
 - ▶ They compromised the software production platform and inserted a backdoor in the code
 - ▶ A compromised update was released to 18000 customers 03-06.2020
 - ▶ US mostly affected, but also 150 victims in Denmark
 - ▶ Attributed to the Russian intelligence service SVR
- ▶ The other attack?
 - ▶ Evidence for parallel attack by a different threat actor



Or to put slightly differently

“For it is no longer a question of ‘if’, but ‘when’ and ‘how often’. I am convinced that there are only two types of companies: those that have been hacked and those that will be. And even they are converging into one category: companies that have been hacked and will be hacked again.”

- FBI Director Mueller, 2012

General Information

Disclaimer: The tools and techniques in this lecture are described for educational purposes only. Using them in real world on targets that don't give you permission to do so can be highly unethical and lead to legal repercussions.

Do not do it!

Learning Goals

► Networks

- Fundamentals
- Application Layer
- Transport Layer
- Network & Link Layer

► Cybersecurity

- Fundamentals
- User Authentication
- Human Factors
- Privacy & Data Protection
- Offensive Security

Guest Lecture

There will be guest lectures. Details will be given, when the date gets closer.

The up-to-date schedule with topics will always be on itslearning

Learning Goals

▶ Networks

- ▶ Fundamentals
- ▶ Application Layer
- ▶ Transport Layer
- ▶ Network & Link Layer

▶ Cybersecurity

- ▶ Fundamentals
- ▶ User Authentication
- ▶ Human Factors
- ▶ Privacy & Data Protection
- ▶ Offensive Security

▶ Humboldt ideals

“An autonomous individual is to be an individual who attains self-determination and responsibility through use of reason.”

- ▶ Be autonomous
- ▶ Be reasonable
- ▶ Be self-determined
- ▶ Take responsibility for your actions

A Note on the Use of Generative AI

- ▶ The choice is yours
- ▶ I recommend not to use it initially, but only after you can judge what the AI gives you
- ▶ AI is likely to make you more efficient at task performance
- ▶ Your skills will not develop if you use it right away
- ▶ We will look at some examples throughout the lecture

Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task[△]

Nataliya Kosmyna ¹
MIT Media Lab
Cambridge, MA

Eugene Hauptmann
MIT
Cambridge, MA

Ye Tong Yuan
Wellesley College
Wellesley, MA

Jessica Situ
MIT
Cambridge, MA

Xian-Hao Liao
Mass. College of Art
and Design (MassArt)
Boston, MA

Ashly Vivian Beresnitzky
MIT
Cambridge, MA

Iris Braunstein
MIT
Cambridge, MA

Pattie Maes
MIT Media Lab
Cambridge, MA

“While LLMs offer immediate convenience, our findings highlight potential cognitive costs. Over four months, LLM users consistently underperformed at neural, linguistic, and behavioral levels.”

“Once men turned their thinking over to machines in the hope that this would set them free. But that only permitted other men with machines to enslave them.”

Frank Herbert, Dune, 1965

Structure of the course

Lecture

- ▶ Each week Tuesday (mostly)
 - ▶ Time → check every time please
 - ▶ Room → check every time please
- ▶ Each week Thursday
 - ▶ Time → check every time please
 - ▶ Room → check every time please

About the lectures

- ▶ The lecture slides will be made available before the lecture on itslearning as a pre-version, so you can take notes on them
 - ▶ Post-versions will include any updates & correct typos I spot
- ▶ All lectures will be recorded in a least effort fashion
 - ▶ Least effort: no dedicated mic, no recording of what happens on the blackboard, in the classroom, etc.
 - ▶ Videos will be available at some point after the lecture, might not be complete
 - ▶ Always check the slide deck, since errata will only be in the slides, not the videos
- ▶ Unless noted on the slides as seen on this one, every slide from the other decks is relevant for the exam

This will not be in the exam

QR code and link for Zoom meeting



<https://syddanskuni.zoom.us/j/61618898702?pwd=Pz53AULdjiqmT0IEJnnWTfzW9pMzzA.1>

Structure of the course

Lecture

- ▶ Each week Tuesday (mostly)
 - ▶ Time → check every time please
 - ▶ Room → check every time please
- ▶ Each week Thursday
 - ▶ Time → check every time please
 - ▶ Room → check every time please

Exercise

- ▶ Each week Wednesday
 - ▶ 14:00 – 16:00
 - ▶ Room → check every time please
- ▶ Each week Friday
 - ▶ 12:00 – 14:00
 - ▶ Room → check every time please

There are exceptions to this, so it is actually best to just check every time

About the exercises

- ▶ The exercise sheets will be made available after the first lecture each week, at the latest on the day of the exercise
- ▶ Exercises are not mandatory, you do not have to hand them in, but doing them is highly recommended
- ▶ There will be solutions put up on itslearning as well

About the exam

- ▶ Exam will include lecture and exercise content
- ▶ Written exam
- ▶ Length: 3h
- ▶ Date: 8th January (tentitatively)
- ▶ Expect at least one question on each of the topics we covered
- ▶ Most questions will be written answers, no multiple choice
- ▶ No aids are allowed!

How you are signed up makes a difference!

DM586

Lecture + written exam
7.5 ECTS

DM572

Lecture + written exam
7.5 ECTS

Assignments
2.5 ECTS

AI510

Lecture + written exam
7.5 ECTS

Innovation
Project
2.5 ECTS

Same as for
computer science
students who have
it attached to
DM585

This side for DM572

Assignments

- ▶ Two assignments to complete for this course worth 2.5 ECTS
 - ▶ One on networks
 - ▶ One on cybersecurity
- ▶ Available on itslearning in the next weeks
- ▶ You have to hand it in individually by 4th December 2026 (AoE)

This side for DM586 & AI510



This side for AI510

Innovation Project

- ▶ You will have to pretend to be a start-up, create a business plan, and develop a product
- ▶ You will find additional time slots for this in week 39-46
- ▶ Since this is a cross-degree course you can also work with students from the other degree

This side for DM586 & DM572



This side for AI510

Innovation Project

- ▶ In week 39 there will be an introduction to this project
- ▶ In weeks 44-46 there will be the mid-term presentation for your project
- ▶ You have to hand it in at the end of the semester
- ▶ You can ask questions to this TA

This side for DM586 & DM572



Some last notes on th schedule

▶ Week 41

▶ There will be no normal lecture

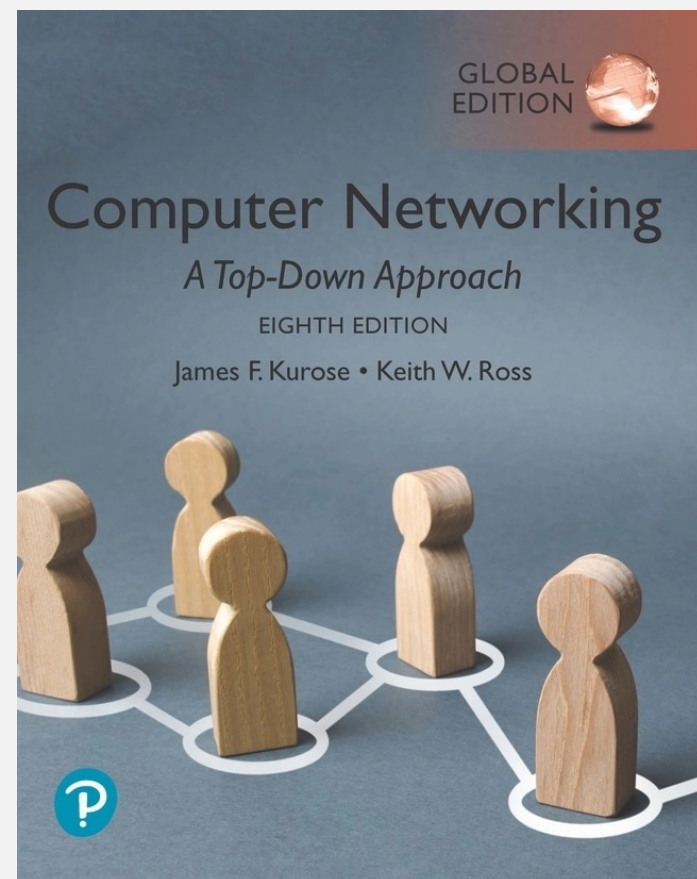
▶ No in-person lectures or exercises in that week

▶ I have to be at a conference to present a paper for a PhD student, since she is on parental leave

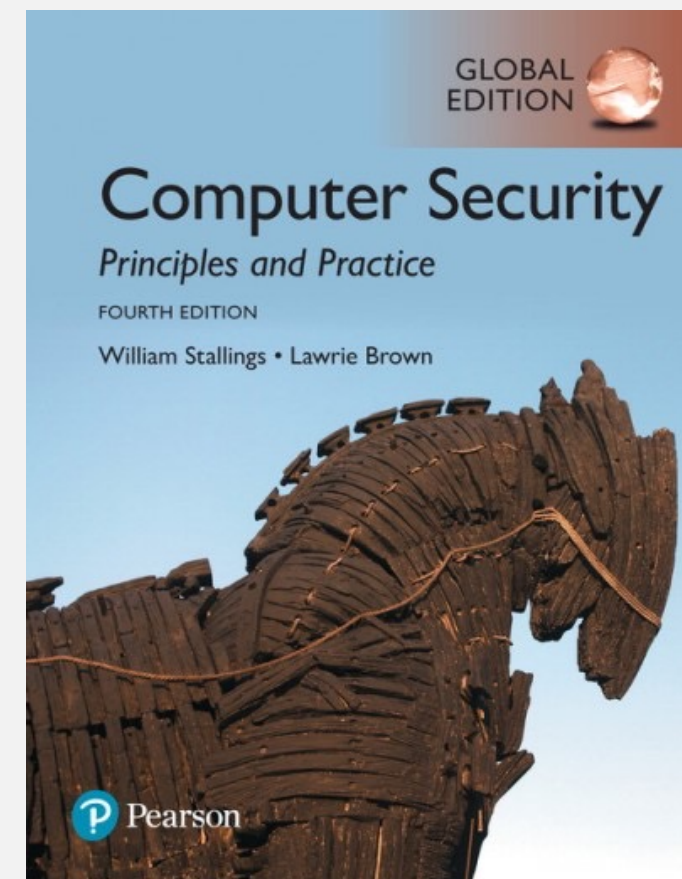
▶ A guest lecture on AI and Security will be provided to you as a video

Literature

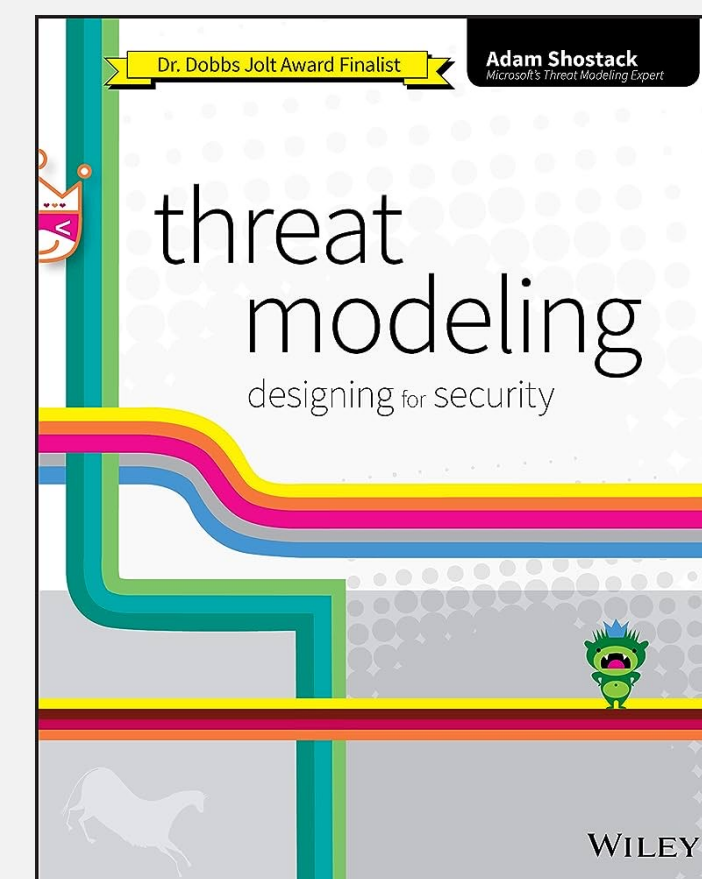
- ▶ You can pass the networking part without the book, but I recommend you get a copy (even an older one will do)
- ▶ For the security part, it is not that easy since there are few (read: none) recent good general-purpose cybersecurity books around



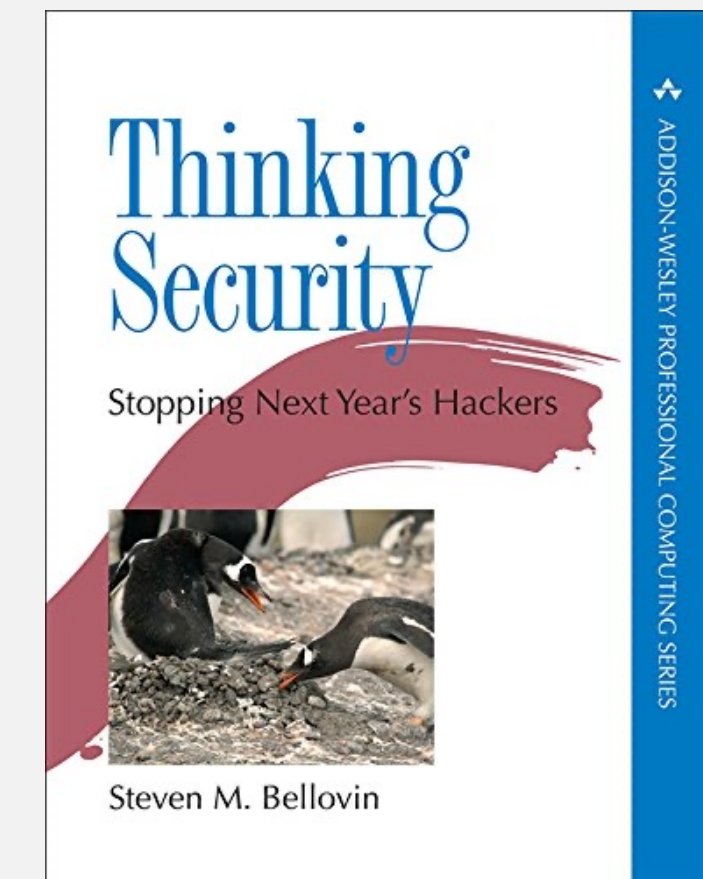
Kurds and Ross,
Computer Networking: A
Top-Down Approach
(8th Edition)



Stallings and Brown,
Computer Security:
Principles and Practice
(4th Edition)



Shostack, Threat
Modelling: Designing for
Security
(1st Edition)



Bellovin, Thinking
Security
(1st Edition)

Any questions so far?